Collections Recovery Optimization: **A Data-Driven Framework**

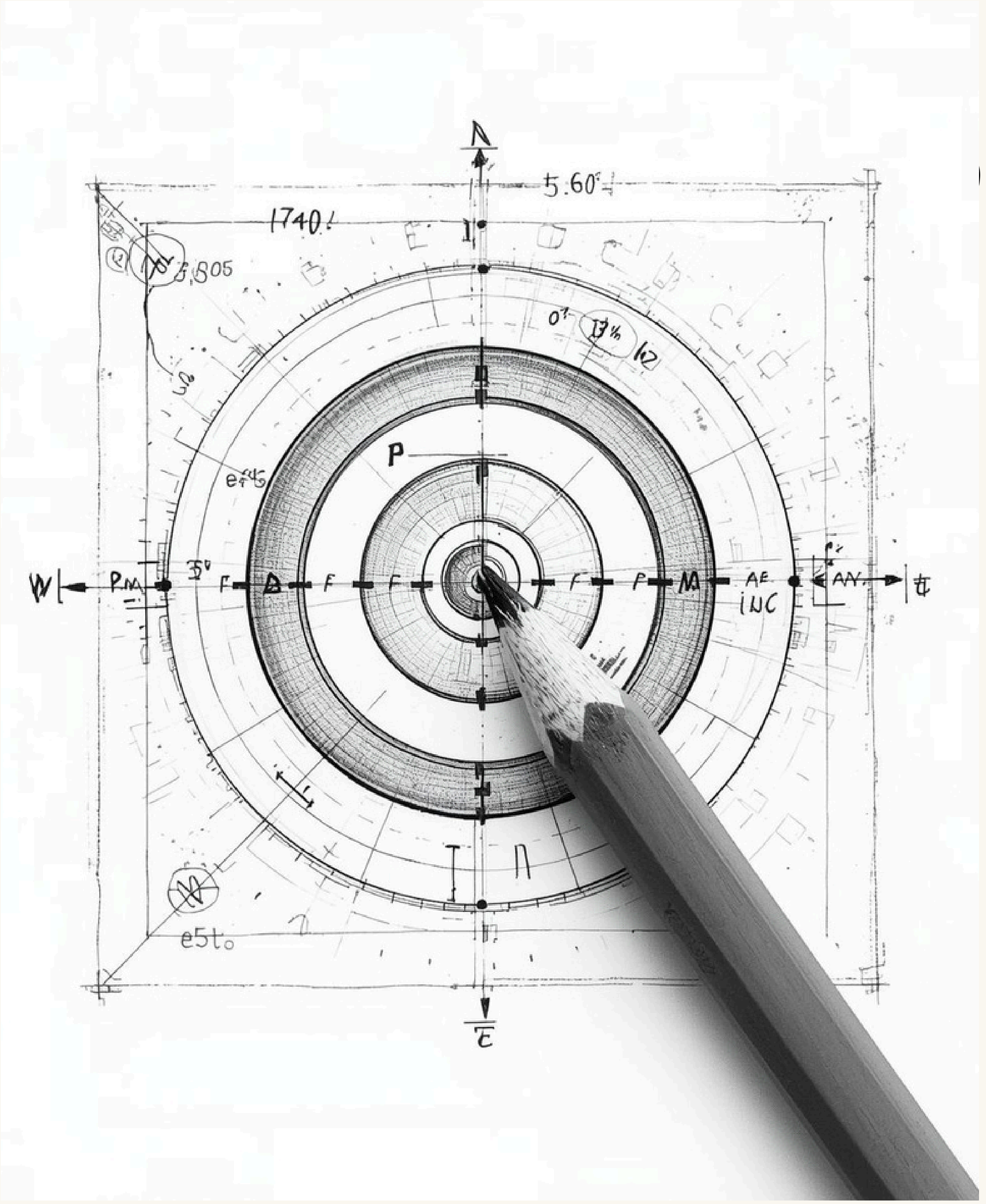
Strategy & Operations Project

Dataset Used:
Lending Club Loans (2007-2018)
2.26M loans | $34B portfolio value

The Collections Problem: Effort Without Strategy

Most collections teams operate with a one-size-fits-all approach:

- Call every overdue account the same number of times
- Use the same channel or one channel (usually phone calls) for everyone
- Send generic messages regardless of customer risk profile
- No A/B testing (channels, timing, personalization etc) to know what actually works



The Result?

- High operational costs (using single channel with not much results)
- Low recovery rates (wrong customers, wrong time, wrong channel)
- Wasted effort on accounts that won't respond

Why This Matters:

This means millions in lost recoveries and inefficient operations

In this project we will whether a data-driven, segmented approach can improve outcomes while reducing costs.

Hypothesis: Smarter Strategy, Better Results

If we segment our customers by risk profile and optimize contact strategy by timing, channel, and messaging, we can improve recovery rates while reducing operational costs.

How We Tested This:

SEGMENTATION
Grouped 2.26M borrowers into risk-based segments (Will Pay/ Might Pay/ Won't Pay)

CONTACT OPTIMIZATION
Analyzed when and how to reach each segment for highest response (time of the day, channel choice)

EXPERIMENTATION
Designed A/B test framework to measure messaging framework

BUSINESS IMPACT
Calculated the increase in revenue and ROI of recommended channel

Dataset: Lending Club loans (2007-2018) | 2,261K loans | $34B portfolio
Analysis Tools: Python(pandas, seaborn, matplotlib), Power Bi

Approach: From Raw Data to Strategy

Analyzed lending data from 2007-2018 to understand what separates good recoveries from bad ones.. Taken into consideration 50,000+ accounts

Step 1: Clean and Structure the Data

- 2.26M loans with payment history, defaults, demographics
- Simplified loan status into: Performing vs. Default
- Focused on recoverable vs. non-recoverable accounts

Step 2: Building Risk Segments

- Used loan grades (A-G) to create three proper segments
- Analyzed & calculated default rates, recovery patterns for each
- Identified which segments responded to collections efforts

Step 3: Created Contact Behavior (Simulation)

- Created 5,000 random contact attempts across times/channels
- Modeled response rates based on segment + contact strategy
- Identified optimal timing and channel combinations

Step 4: Designed Experiment

- Built A/B test comparing generic vs. personalized messaging
- Measured the significance and business impact
- Calculated projected ROI and payback period

NOTE:

This analysis uses historical data as a proof-of-concept.
Real deployment would require client-specific data and testing.

Finding 1: Not All Customers Are Equal

Grouped borrowers into three risk-based segments:/
distinct buckets for analysis

Will Pay (Grades A-B)

- High likelihood of repayment (48% of portfolio)
- Low default rate (8%)
- Best response to light-touch channel

Might Pay (Grades C-D)

- Sensitive to strategy & changes (43% of portfolio)
- Moderate default rate (15%)
- Needs optimized contact approach

Won't Pay (Grades E-G)

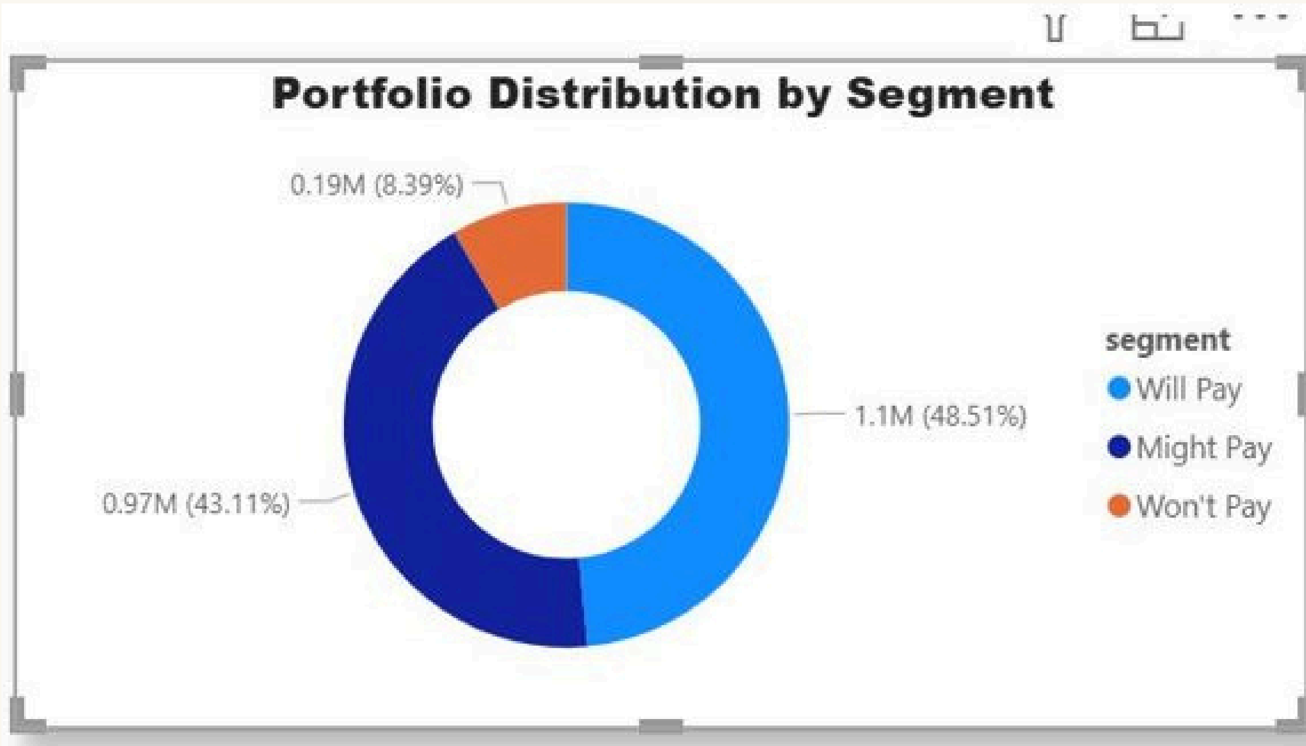
- Low recovery likelihood (9% of portfolio)
- High default rate (28%)
- Expensive to pursue, low return

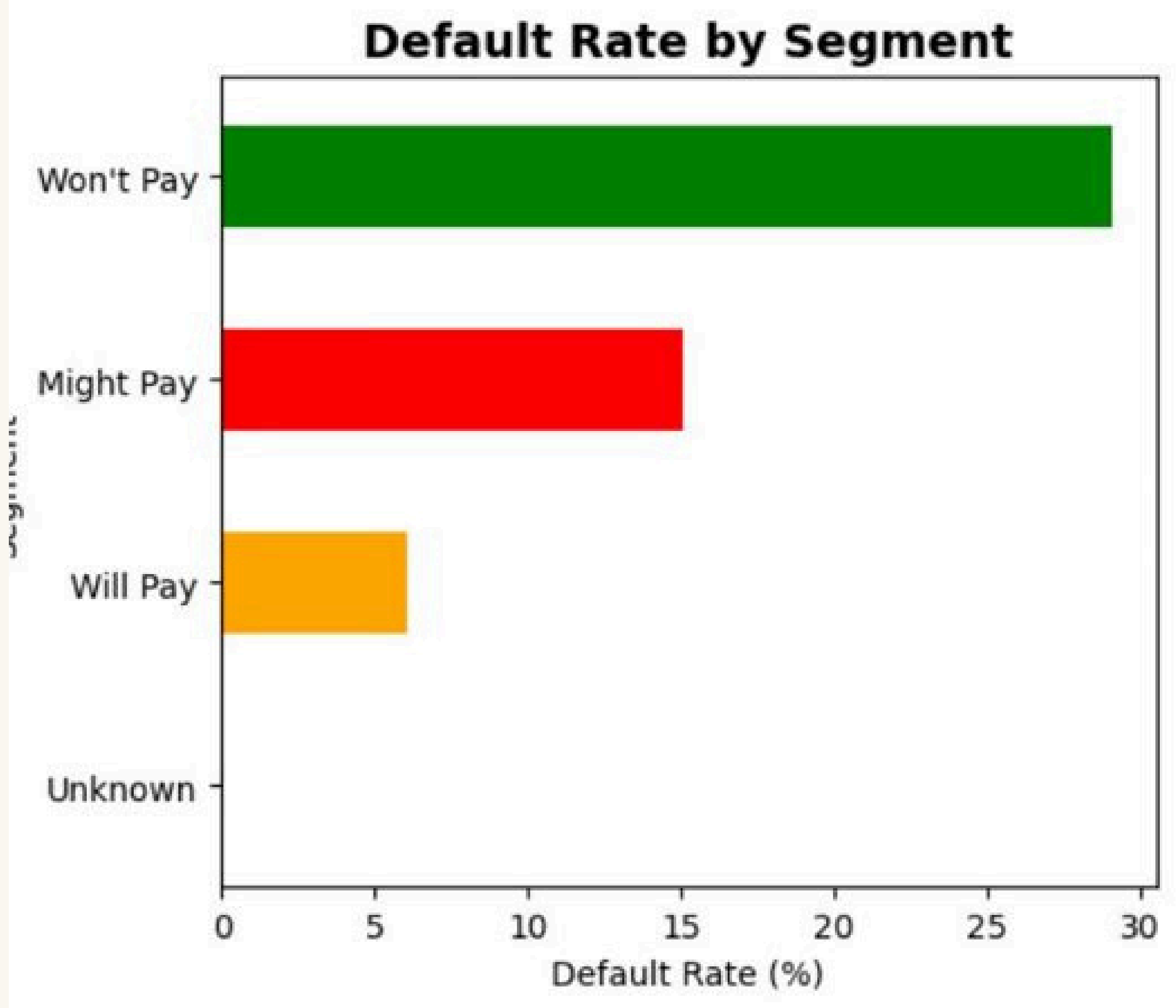
Key Insight:

A small portion of high-risk accounts drive most defaults, while nearly half the portfolio is highly recoverable with the right approach

Why This Matters

- Prevents wasting resources on low-intent borrowers
- Unlocks higher ROI by focusing on "Might Pay" segment
- Enable us to build different strategies for each instead of treating everyone the same





FINDING 2 - TIMING AND CHANNEL MATTER

Created 5,000 contact attempts (random generation) to understand how response rates change based on timing, channel, and segment.

Timing Insight: Evening(Winner)

- 6-8 PM shows 55%+ response rates
- Morning (9-11 AM) drops to 33%
- Implication: Shift contact windows to evening hours

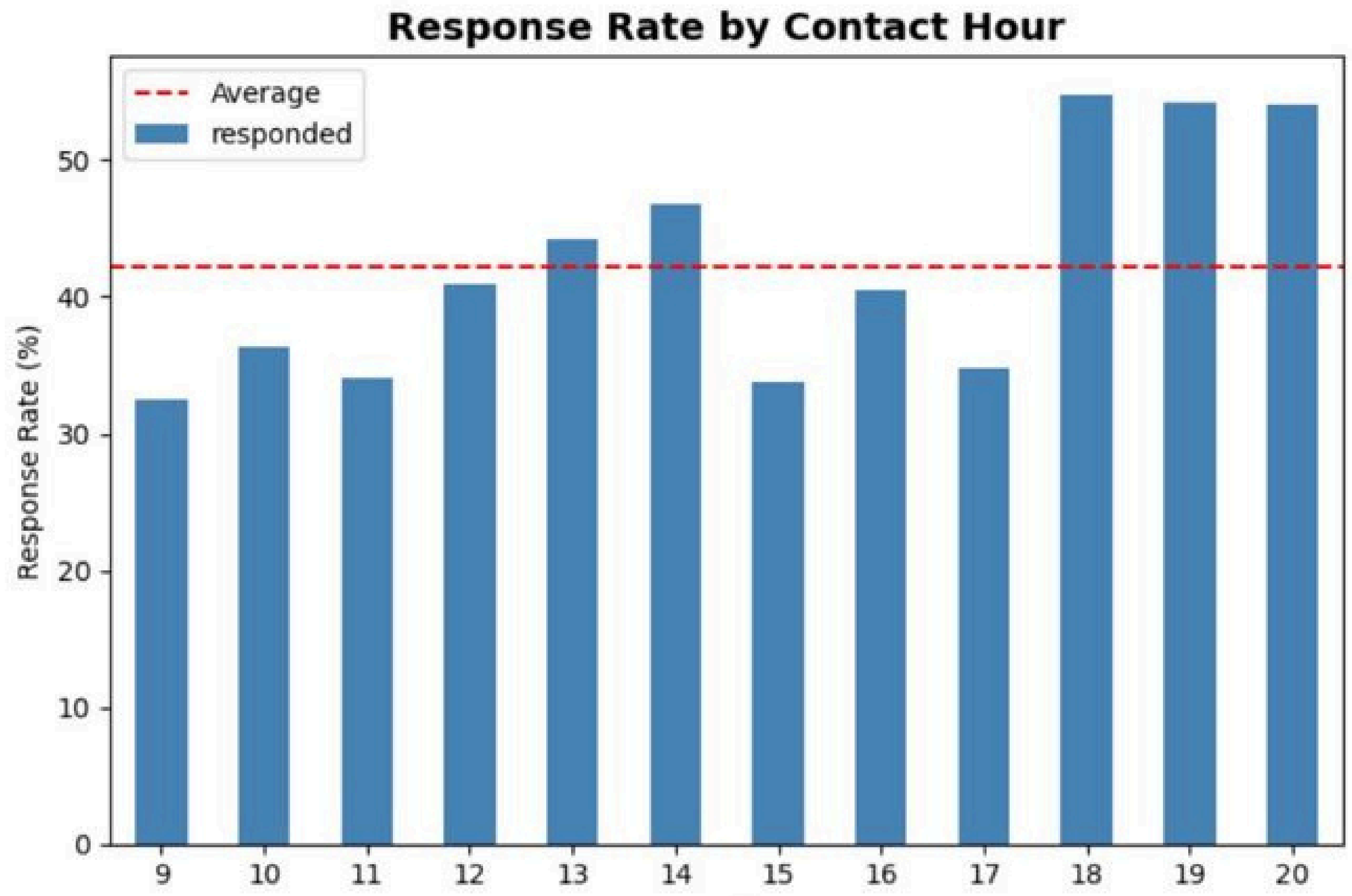
Channel Insight: One Size Do NOT Fit All

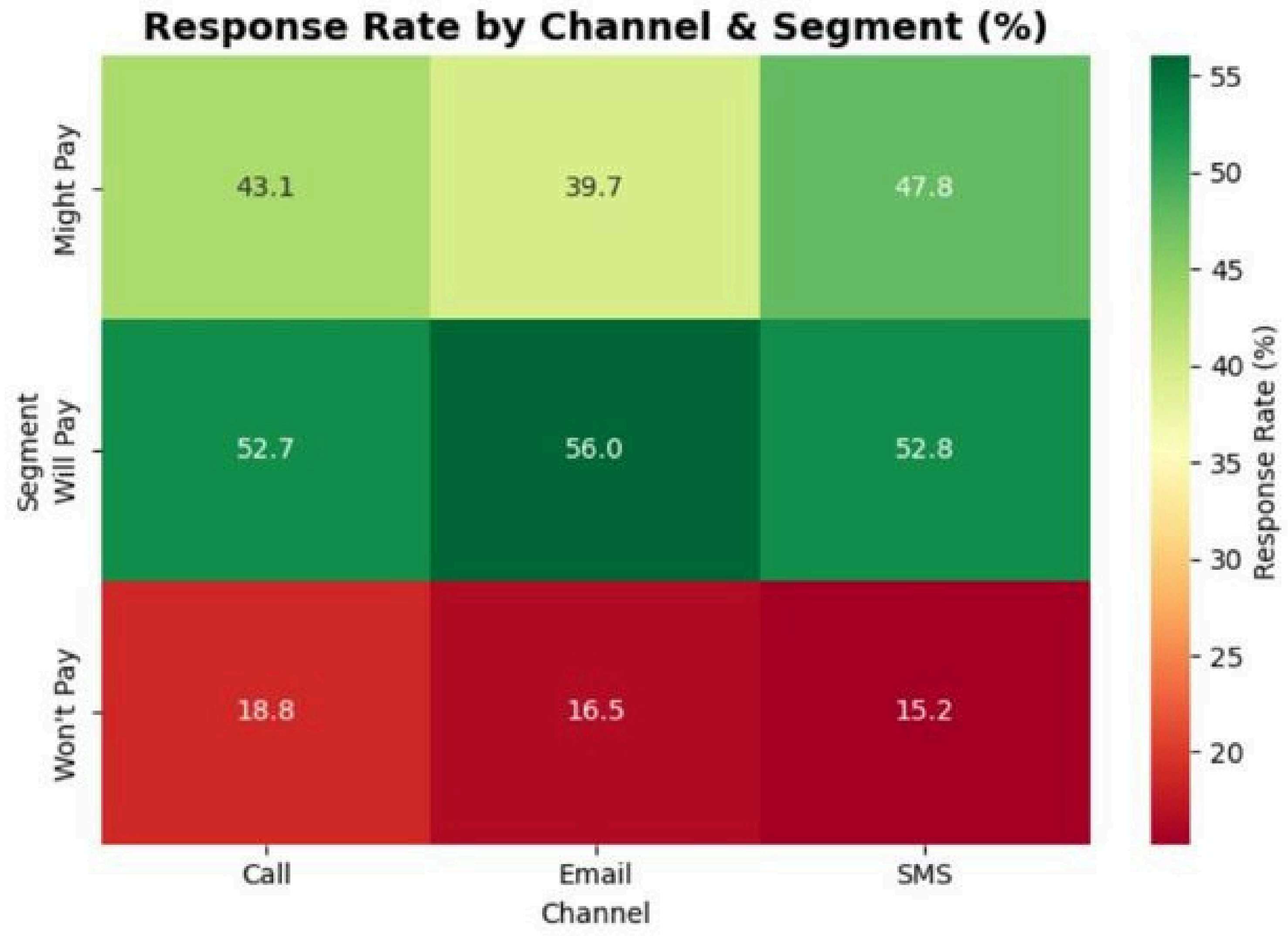
- Will Pay segment: Email performs best (56% response)
  - Lower cost, they're already engaged
- Might Pay segment: SMS outperforms (53% response)
  - Quick, not-intrusive, immediate & action-oriented
- Won't Pay segment: Calls perform relatively better (19%)
  - Direct intervention, but still low ROI

Cost/Financial Impact

- Email: $0.10 per contact
- SMS: $0.50 per contact
- Call: $2.00 per contact

Multichannel/ channel routing as per the segment could reduce costs by 60% while maintaining or improving response rates.





Finding 3: Personalized Messaging Drives 19.6% Lift

Designed an A/B test to measure the impact of personalized messaging:

Test Design:

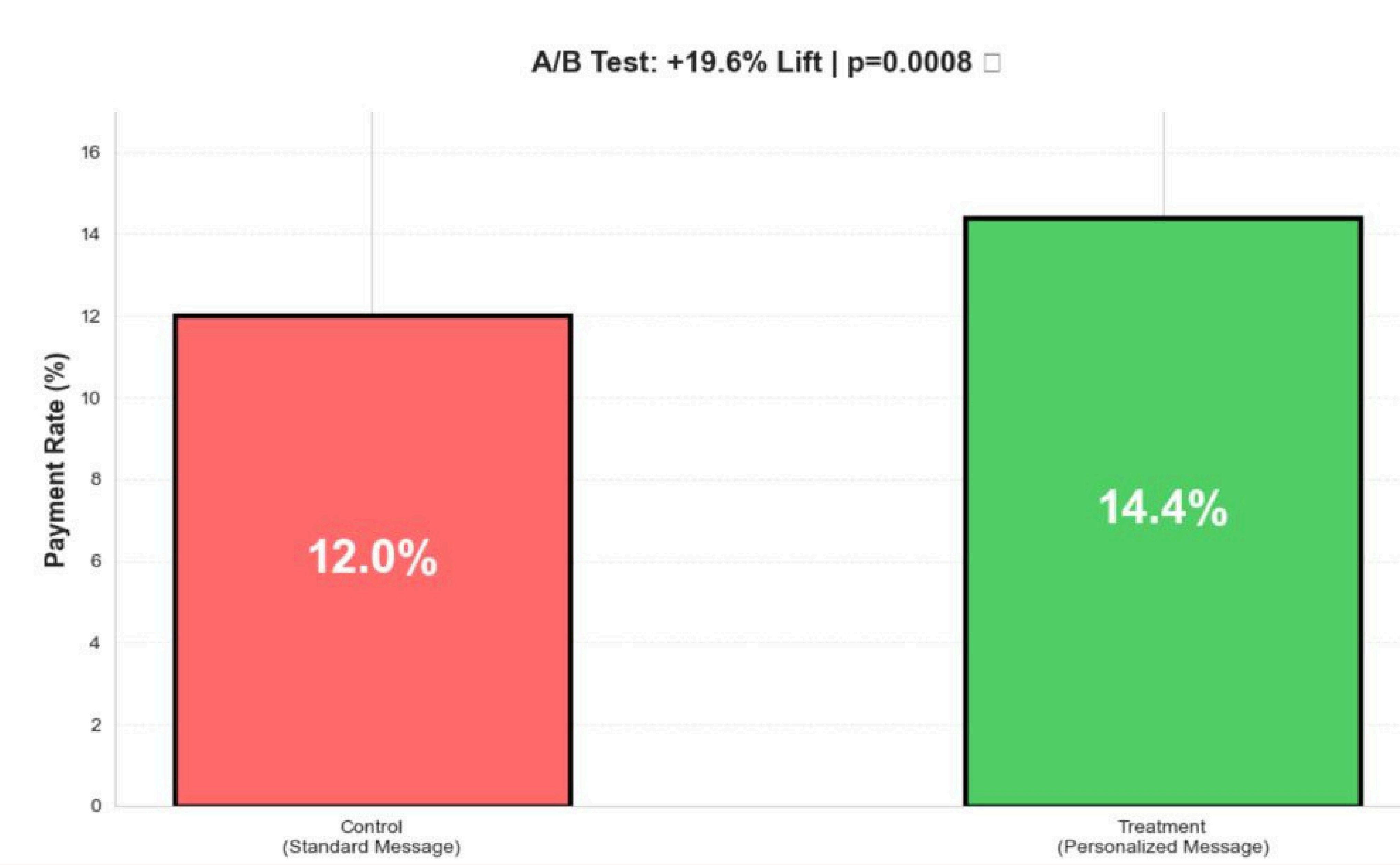
- Control Group (5,000): Generic message
  - "Your payment is past due. Please contact us."
- Treatment Group (5,000): Segment-aware personalized message
  - Example for "Might Pay": "We understand things happen. Let's find a payment plan that works for you."

Results:

- Control: 12.0% payment rate
- Treatment: 14.5% payment rate
- Lift: +19.6% improvement (Significance p = 0.0008)

What This Shows:

Small changes in messaging, when personalized as per the segment behavior, creates a measurable improvement.
This validates our hypothesis that that continuous experimentation (A/B testing) and not one size fits all is the key for optimizing collection effeciency



NOTE:

This is a simulated or random data generated proof-of-concept. Real deployment would require live testing with client data through companies platform.

Business Impact: $1.53M Annual Uplift + 60% Cost Reduction

Projected Impact (100,000 annual overdue accounts)

Revenue Impact:

- Current recovery: 12% × 100K × $500 = $6.0M/year
- Improved recovery: 14.5% × 100K × $500 = $7.25M/year
- Incremental revenue: +$1.25M/year

Cost Impact:

- Current: 100K accounts × 8.5 calls × $2 = $1.7M/year
- Optimized: Smart channel routing = $680K/year
- Cost savings: -$1.02M/year

Combined Annual Benefit: $2.27M

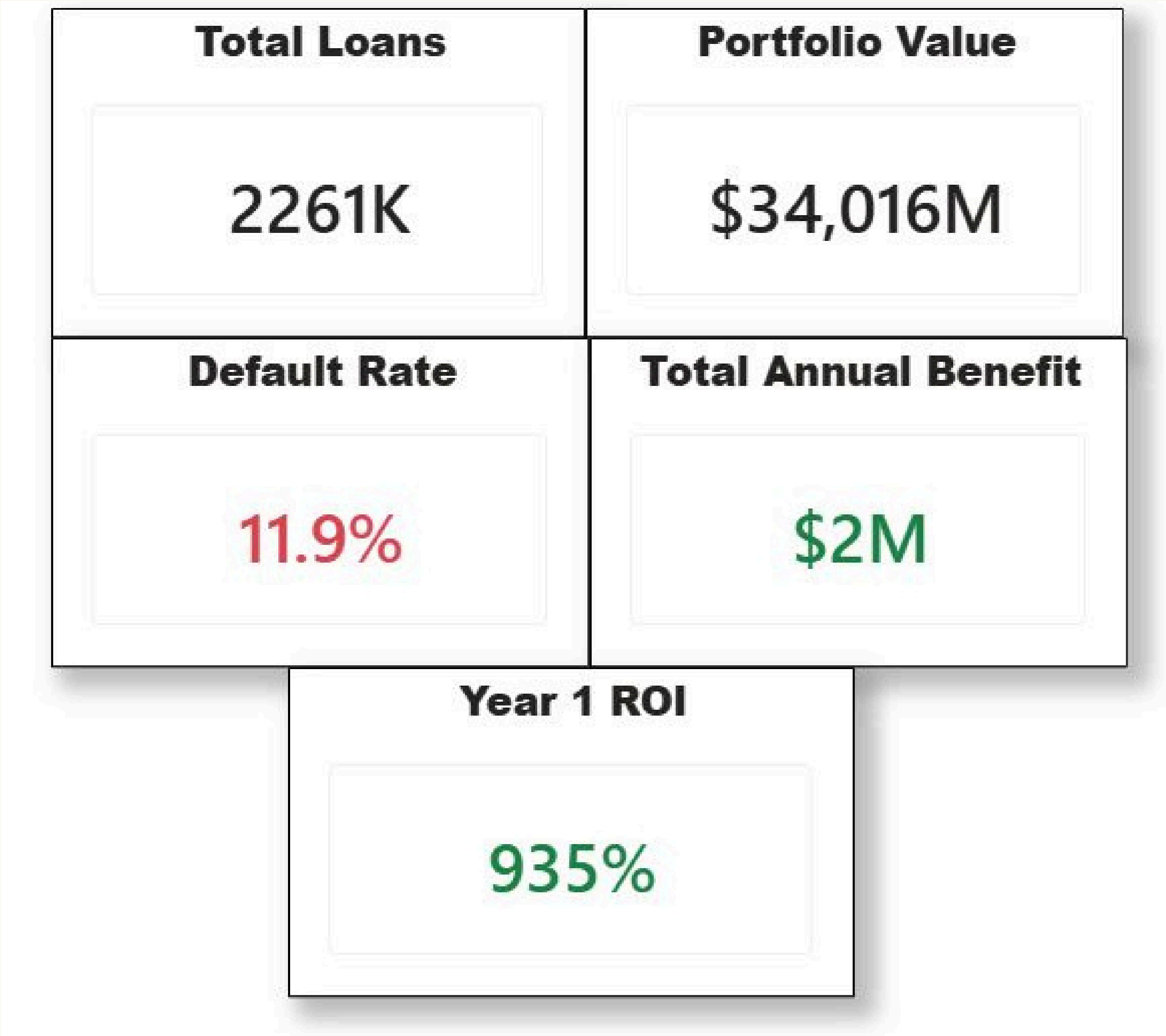
ROI:

- Year 1: 935% ROI
- Payback: 1.1 months
- Years 2+: 4,440% ROI

Total Annual Benefit
$2.27M

Lift In Recovery
19.6%

Payback Period
1.1 months



Why This Matters:

This framework is exactly where at scale helping clients like banks and NBFCs deploy segment-aware, channel-optimized strategies that drive measurable recovery improvements.

Implementation Roadmap

This analysis is a starting point. Here's how I'd scale it in a real Strategy & Operations role at company:

Phase 1: Validate (Months 1-2)

- Work with 2-3 pilot clients to test segmentation logic
- Validate contact optimization patterns with real data
- Refine segment definitions based on client portfolios

Phase 2: Experiment (Months 3-4)

- Launch A/B tests on messaging and channel strategies
- Measure lift across different client types (banks vs NBFCs
- Build statistical confidence before scaling

Phase 3: Operationalize (Months 5-6)

- Build automated decisioning rules into PIE platform
- Create client-facing dashboards for performance tracking
- Train client teams on strategy execution

Phase 4: Continuous Improvement (Ongoing)

- Weekly performance reviews with clients
- Monthly strategy refinements based on test results
- Quarterly portfolio reviews to adapt to market changes

Success Factors:

- Close partnership with companies data science team
- Strong client relationships (CXO-level engagement)
- Disciplined experimentation + rapid iteration
- Clear measurement framework for every change

What We Learned: Four Insights

1

Collections Is Not Just a Risk Problem

Default rates tell you who might not pay, but not how to reach them.

When and how you contact matters as much as who you contact. Timing, channel, and messaging are levers, not just operational details.

2

Segmentation Prevents Wasted Effort

Treating all over due accounts the same burns resources on low-intent borrowers while under-serving recoverable ones.

Segment-aware strategies let you focus effort where it converts.

3

Small Changes, Measurable Impact

A 19.6% lift from personalized messaging shows that incremental improvements not overhauls drive results.

Continuous experimentation beats one size fits all playbooks.

4

Cost and Recovery Are Not Trade-Offs

Smart channel routing reduces costs by 60% while maintaining or improving recovery rates.

The right strategy improves both efficiency and effectiveness.

The End